overview

the module, the project, and where everything belongs · prof. dr. nicolas meseth

1 why this module
2 what you will learn
3 the project
4 experiment: you are the receiver
5 1 + 4 challenges
6 how we work
7 this semester

quick poll

have you ever written a program?

yes, quite a lot

a little, at school or for fun

never



live.hsos.uber.space/q

quick poll

have you ever written a program?

yes, quite a lot

a little, at school or for fun

never



live.hsos.uber.space/q

quick poll

how often do you use ai chats like ChatGPT?

every day

every week

now and then

never

live.hsos.uber.space/q

quick poll

how often do you use ai chats like ChatGPT?

- every day
- every week
- now and then
- never

live.hsos.uber.space/q

quick poll

have you ever worked with an AI agent, like Claude Code, Codex or OpenCode?

yes

i have heard of them

no, never heard of them

live.hsos.uber.space/q

quick poll

have you ever worked with an AI agent, like Claude Code, Codex or OpenCode?

yes

i have heard of them

no, never heard of them

live.hsos.uber.space/q

quick poll

how well do you know what happens inside a computer?

very well

roughly

hardly

not at all



live.hsos.uber.space/q

quick poll

how well do you know what happens inside a computer?

very well

roughly

hardly

not at all

live.hsos.uber.space/q

quick quiz

how many different values can one byte hold?

8

16

256

no idea



live.hsos.uber.space/q

quick quiz

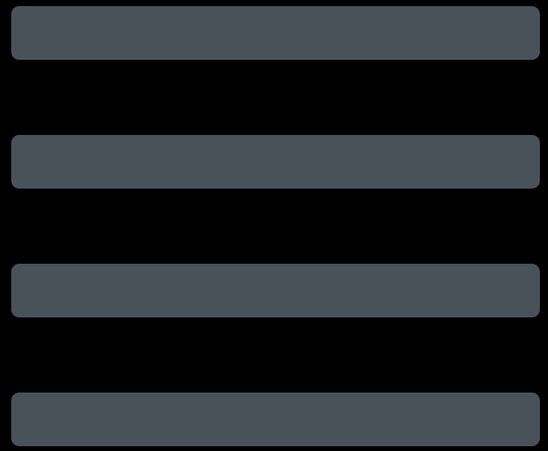
how many different values can one byte hold?

8

16

256

no idea



live.hsos.uber.space/q

quick quiz

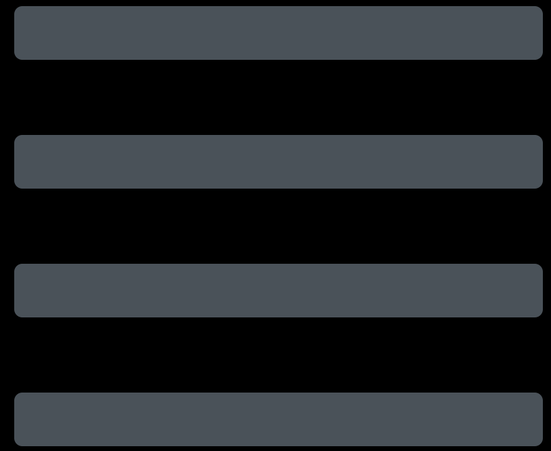
how many different values can one byte hold?

8

16

256

no idea



live.hsos.uber.space/q

quick quiz

a photo on your phone: roughly how many numbers are stored for it?

a few hundred

a few thousand

tens of millions

no idea



live.hsos.uber.space/q

quick quiz

a photo on your phone: roughly how many numbers are stored for it?

- a few hundred
- a few thousand
- tens of millions
- no idea

live.hsos.uber.space/q

quick quiz

a photo on your phone: roughly how many numbers are stored for it?

- a few hundred
- a few thousand
- tens of millions
- no idea

live.hsos.uber.space/q

quick quiz

at the lowest level, what does a computer work with?



live.hsos.uber.space/q

words

pictures and sounds

numbers: zeros and ones

no idea

quick quiz

at the lowest level, what does a computer work with?

words

pictures and sounds

numbers: zeros and ones

no idea

live.hsos.uber.space/q

quick quiz

at the lowest level, what does a computer work with?

words

pictures and sounds

numbers: zeros and ones

no idea

live.hsos.uber.space/q

quick quiz

you switch off your computer. what happens to what is in its ram?

it stays there

it is lost

no idea



live.hsos.uber.space/q

quick quiz

you switch off your computer. what happens to what is in its ram?

it stays there

it is lost

no idea

live.hsos.uber.space/q

quick quiz

you switch off your computer. what happens to what is in its ram?

it stays there

it is lost

no idea





live.hsos.uber.space/q

quick poll

how important will it be for your career to understand AI and to work well with it?

essential

very important

somewhat important

not important



live.hsos.uber.space/q

quick poll

how important will it be for your career to understand AI and to work well with it?

essential

very important

somewhat important

not important

live.hsos.uber.space/q

part 1

why this module

AI answers any question in seconds.

AI answers any question in seconds.
it writes, summarises, translates and explains. it seems to know everything.

AI answers any question in seconds.
it writes, summarises, translates and explains. it seems to know everything.
or does it?

AI answers any question in seconds.
it writes, summarises, translates and explains. it seems to know everything.
or does it?

so why learn anything at all?

the same question, fifty years ago:

everyone has a calculator. why still learn maths?



image: AI-generated (gpt-image-2.5-flare)

to write the problem down so that the calculator can solve it.

to write the problem down so that the calculator can solve it.
to notice when 4,000 comes out and 40 would be right.

to write the problem down so that the calculator can solve it.
to notice when 4,000 comes out and 40 would be right.
to know what it cannot calculate at all.

to write the problem down so that the calculator can solve it.
to notice when 4,000 comes out and 40 would be right.
to know what it cannot calculate at all.

the same holds for computers, and for AI.

a computer is just a bigger calculator.

it turns numbers into numbers, billions of times a second. to judge what comes out, you have to understand what it does.

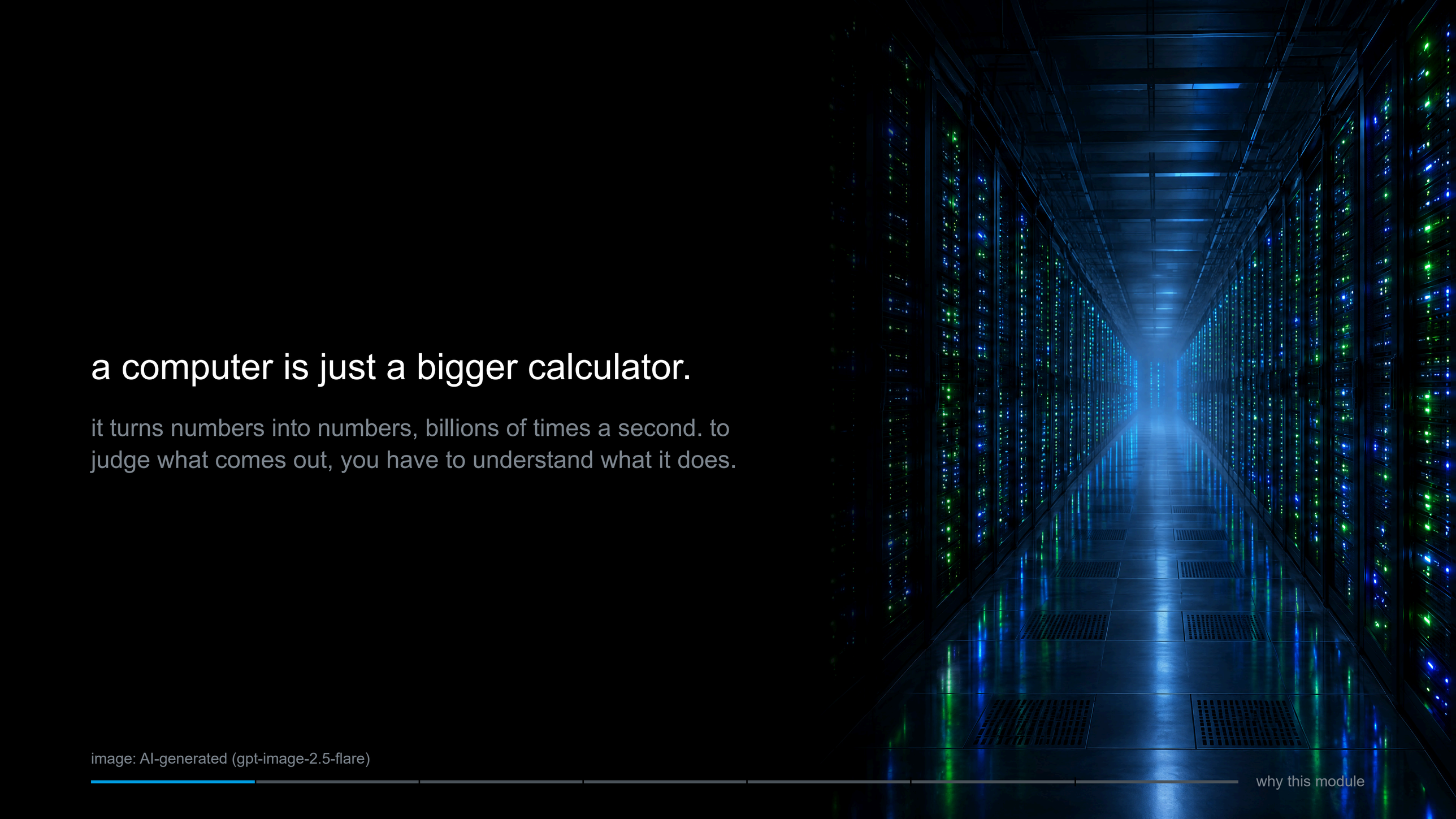


image: AI-generated (gpt-image-2.5-flare)

pilots still learn how a plane flies.

the better the autopilot, the rarer the moment a human takes over, and the more that moment depends on understanding.



image: AI-generated (gpt-image-2.5-flare)

it answers anything in seconds.



image: AI-generated (gpt-image-2.5-flare, edited with gpt-image-2.5-sunburst)

it answers anything in seconds.
it sounds right, even when it is wrong.



image: AI-generated (gpt-image-2.5-flare, edited with gpt-image-2.5-sunburst)

it answers anything in seconds.
it sounds right, even when it is wrong.
it cannot check its answer against the real world.



image: AI-generated (gpt-image-2.5-flare, edited with gpt-image-2.5-sunburst)

it answers anything in seconds.
it sounds right, even when it is wrong.
it cannot check its answer against the real world.

you learn to work with it: cut, check, measure.



image: AI-generated (gpt-image-2.5-flare, edited with gpt-image-2.5-sunburst)

AI

is software: a program, like any other.

and to understand AI

AI

is software: a program, like any other.

software

runs on a computer.

and to understand AI

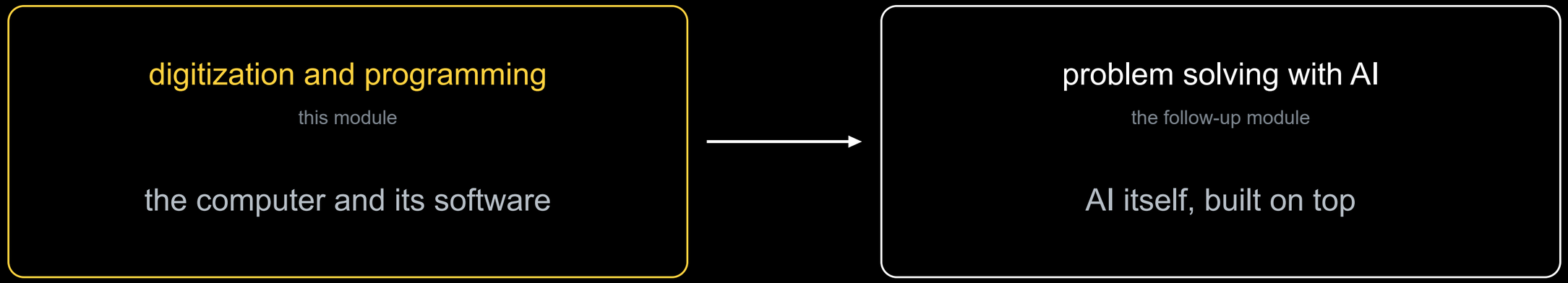
AI — is software: a program, like any other.

software — runs on a computer.

computer — turns numbers into numbers, billions of times a second.

AI	is software: a program, like any other.
software	runs on a computer.
computer	turns numbers into numbers, billions of times a second.

to understand AI, you first understand the computer and its software.



this module is the foundation. the next one builds on it.

would you like to have a better understanding of computers?

yes, definitely

a little

not really



live.hsos.uber.space/q

would you like to have a better understanding of computers?

yes, definitely

a little

not really

live.hsos.uber.space/q

part 2

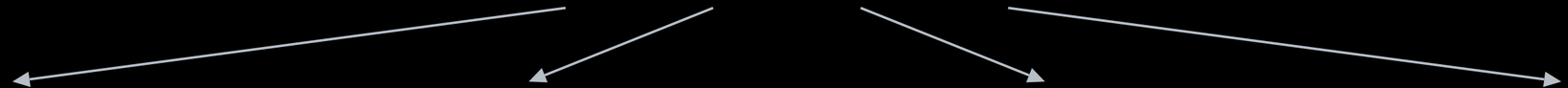
what you will learn

how do computers solve complex problems?

how do computers solve complex problems?

- how do computers represent information?
- how do computers store information?
- how do computers transfer information?
- how do computers process information?

how do computers solve complex problems?



how do computers represent information?

`making bits`

how do computers store information?

`storing bits`

how do computers transfer information?

`transferring bits`

how do computers process information?

`processing bits`

how do computers solve complex problems?

how do computers represent information?

- making bits
- analog and digital
- symbols and information
- number systems
- code systems

how do computers store information?

- storing bits
- memory and storage

how do computers transfer information?

- transferring bits
- signal and noise
- sampling and sync
- protocols
- errors and redundancy
- throughput and limits

how do computers process information?

- processing bits
- logic and arithmetic
- compression
- encryption

cutting problems | input, processing, output | algorithms and programs | measuring and experimenting | abstraction and layers

how do computers solve complex problems?

how do computers represent information?

- making bits
- analog and digital
- symbols and information
- number systems
- code systems

how do computers store information?

- storing bits
- memory and storage

how do computers transfer information?

- transferring bits
- signal and noise
- sampling and sync
- protocols
- errors and redundancy
- throughput and limits

how do computers process information?

- processing bits
- logic and arithmetic
- compression
- encryption

the map lives on the website

LiFi Project

Concepts > Overview

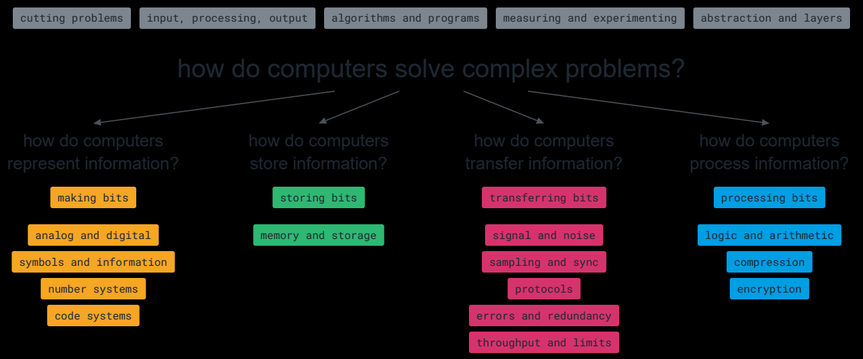
Overview

The module, the project, and where everything belongs

This page is the map. It shows how the module is put together: the one question behind it, the four questions underneath, the eighteen concepts and their five colours, and the five challenges of the project with the concepts each of them needs. You will see this map again and again during the semester, on the slides and on this page, each time with the part lit up that we are working on. If you ever lose track of where you are, come back here.

One question, four questions

The module has one big question: **how do computers solve complex problems?** To answer it you need to understand what a computer does with information, and there are four things it does. It represents information, it stores it, it transfers it, and it processes it.



the lifi project

part 3

the project

why a project, and not just lectures?



image: AI-generated from photos of the real device (gpt-image-2.5-sunburst)



learn by doing

concepts stick when you have built them, broken them and fixed them yourself.



learn by doing

concepts stick when you have built them, broken them and fixed them yourself.



problems first

every problem meets you on your own device before it gets a name in class.



learn by doing

concepts stick when you have built them, broken them and fixed them yourself.



problems first

every problem meets you on your own device before it gets a name in class.



reality answers

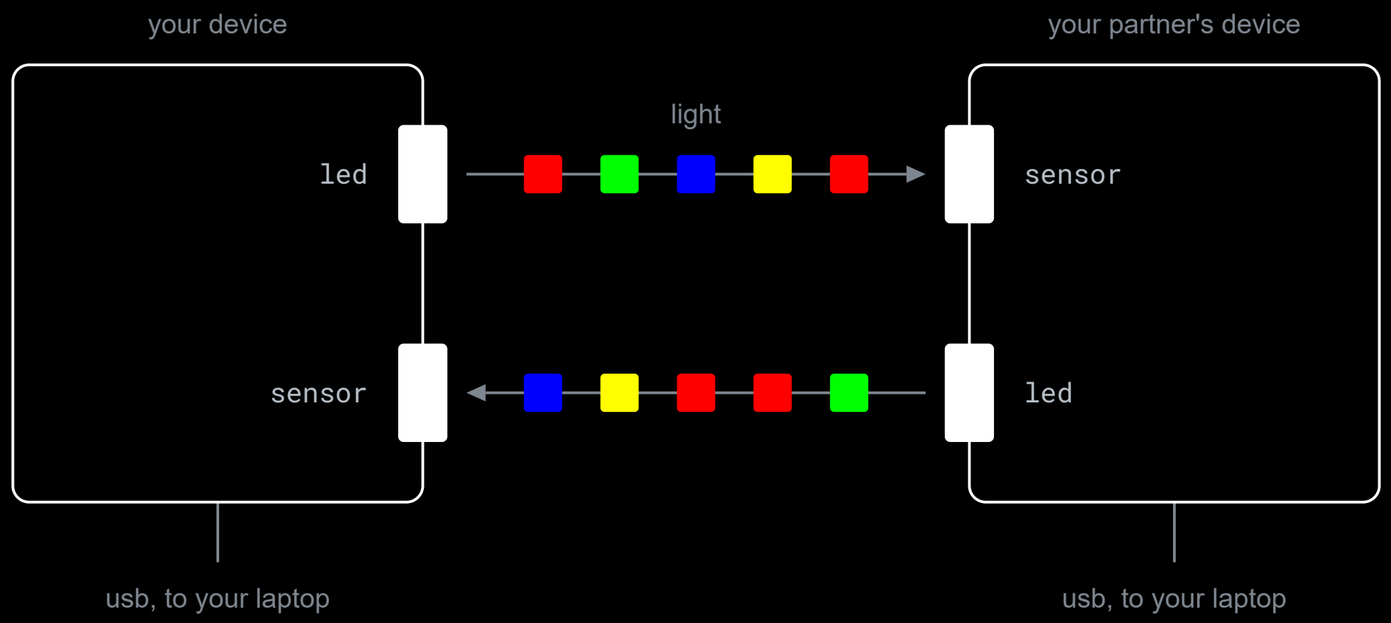
a device does not care how good an answer sounds. it works, or it does not.

there is a photo on this laptop. how does it get to the other one?



image: AI-generated (gpt-image-2.5-flare)





a lamp and an eye on the front of each device. two devices face each other, so light travels both ways.

everything interesting in a computer is invisible.

this project makes it visible: a few times a second, on your desk, where you can explore it with your own hands.

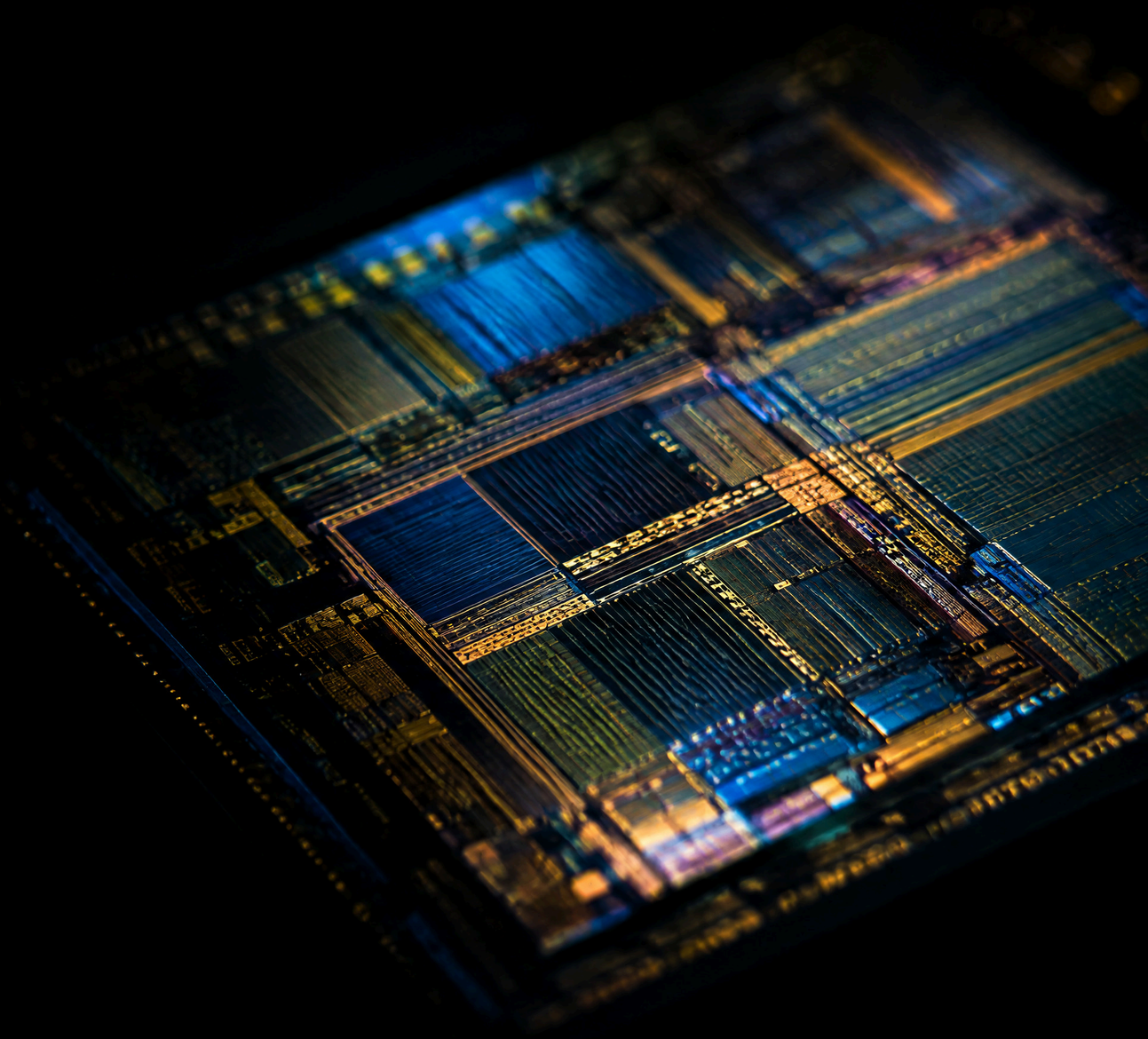


image: AI-generated (gpt-image-2.5-flare)



a photo on your laptop

`memory and storage`

the journey of one photo



a photo on your laptop

becomes bytes

`memory and storage`

`number systems` `code systems`

the journey of one photo



a photo on your laptop

becomes bytes

is packed into a frame with a checksum

`memory and storage`

`number systems` `code systems`

`protocols` `errors and redundancy` `compression` `encryption`

the journey of one photo



a photo on your laptop

becomes bytes

is packed into a frame with a checksum

bytes become colours

memory and storage

number systems · code systems

protocols · errors and redundancy · compression · encryption

symbols and information · analog and digital

the journey of one photo



a photo on your laptop

becomes bytes

is packed into a frame with a checksum

bytes become colours

travels as light

`memory and storage`

`number systems` `code systems`

`protocols` `errors and redundancy` `compression` `encryption`

`symbols and information` `analog and digital`

`signal and noise` `sampling and sync` `throughput and limits`

the journey of one photo



a photo on your laptop — `memory and storage`

becomes bytes — `number systems` `code systems`

is packed into a frame with a checksum — `protocols` `errors and redundancy` `compression` `encryption`

bytes become colours — `symbols and information` `analog and digital`

travels as light — `signal and noise` `sampling and sync` `throughput and limits`

arrives, is checked, becomes a photo again — `logic and arithmetic`

every stop on the way is a concept of this module.

part 4

experiment: you are the receiver

take a pen and a piece of paper.
the screen is the lamp. you are the sensor.
write down every symbol you see, using the alphabet at the bottom.

round 1: four colours, slowly



alphabet r g b y

round 1: four colours, slowly



sent `g r y b r y g b`

alphabet r g b y

round 2: eight blues



alphabet 1 2 3 4 5 6 7 8

round 2: eight blues



sent 3 7 1 5 8 2 6 4

alphabet 1 2 3 4 5 6 7 8

round 3: faster

alphabet r g b y

round 3: faster



sent r y g g b r y b g r b y

alphabet r g b y

round 4: where is the message?



alphabet r g b y

round 4: where is the message?



sent · · · · · · · g b r y g r · · · · ·

alphabet r g b y

round 5: did everything arrive?



alphabet r g b y

round 5: did everything arrive?



sent y g r b g y b r

alphabet r g b y

what went wrong?

too many colours to tell apart

too fast. was that one or two?

where does the message start?

did everything arrive?

what went wrong?

too many colours to tell apart → challenge 1: the alphabet

too fast. was that one or two? → challenge 2: the word

where does the message start? → challenge 3: the listener

did everything arrive? → challenge 4: the packet

part 5

1 + 4 challenges

one new difficulty at a time

0	the spark	can you bring your device to life?
1	the alphabet	how many signals can your receiver tell apart?
2	the word	can you send a whole word, and how fast?
3	the listener	can the receiver find the start on its own?
4	the packet	can you send a real file, correct and fast?

each challenge adds exactly one new difficulty. everything else stays as you know it.

what every challenge produces



measurement logs

what did you change, what did you keep the same, and what came out?



a description of your solution

enough for another team to rebuild it. from challenge 3 on, a full protocol specification.



a log of mistakes

where the assistant suggested something, and a measurement proved it wrong.

nothing of this is handed in or graded. it is for your team.

part 6

how we work



two students, two devices

one device each, so you own both ends of the link. split the work, and swap now and then.



two students, two devices

one device each, so you own both ends of the link. split the work, and swap now and then.



the lamp has the last word

your assistant knows nothing about your device. measure before, measure after, and note where it was wrong.



two students, two devices

one device each, so you own both ends of the link. split the work, and swap now and then.



the lamp has the last word

your assistant knows nothing about your device. measure before, measure after, and note where it was wrong.



try the demonstrators

a small toy in the browser for almost every concept: switch a byte, break a file, add noise.

part 7

this semester

fourteen mondays

1 **kick-off: overview & setup**
devices, setup, challenge 0 starts

2 **cutting problems, algorithms and programs**
challenge 0: your first program

3 **measuring and experimenting**
challenge 0 check

4 **analog and digital, symbols and information**
challenge 1 starts

5 **signal and noise**
🏆 competition: challenge 1

6 **number systems and code systems**
challenge 2 starts

7 **sampling and synchronization**
challenge 2: faster

8 **throughput and limits**
🏆 competition: challenge 2

9 **protocols**
challenge 3 starts

10 **logic and arithmetic**
interop test

11 **memory and storage**
🏆 competition: challenge 3

12 **errors and redundancy**
standardisation session, challenge 4 starts

13 **compression**
challenge 4: faster and shorter

14 **abstraction and layers**
🏆 final: challenge 4

this semester

a typical monday, 15:00 to 18:15

15:00	look back: where are you stuck?
15:30	input: one concept
16:00	experiment, all together
16:30	break
16:45	workshop
18:00	outlook and your task for the week

every monday, the longest block is yours: you build.

this semester

every monday



room HD0001

mondays, 15:00 to 18:15.



laptop and device

bring both, every time. charge the laptop before you come.



attendance list

sign it every week. it is for statistics, not for grades.

this semester



60 minutes

multiple choice on the concepts, everyone on their own. the project is where you meet them.



60 minutes

multiple choice on the concepts, everyone on their own. the project is where you meet them.



+5 % bonus

if challenges 1 to 4 are all accepted. all or nothing.

1 2 3 4



60 minutes

multiple choice on the concepts, everyone on their own. the project is where you meet them.



+5 % bonus

if challenges 1 to 4 are all accepted. all or nothing.

1 2 3 4



practise

with the questions on every concept page. same format as the exam.

where to find what

`docs.lifi-project.de`	everything to read: concepts, challenges, setup
`lms.hs-osnabrueck.de`	announcements, changes, the schedule
your course folder	your assistant and your code
`lifi.hsos.uber.space/cockpit`	your link, live
`demos.lifi-project.de`	the demonstrators

how do you feel about this module?



live.hsos.uber.space/q

looking forward to it

curious

a bit worried

worried

how do you feel about this module?

looking forward to it

curious

a bit worried

worried



live.hsos.uber.space/q

by the end of term,
a picture crosses that gap.

and along the way,
more than one light will go on.



image: AI-generated (gpt-image-2.5-flare)